

An inductive-recursive universe of searchable ordinals

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The problem addressed here

Given a type X and $p: X \rightarrow 2$,

$$2 = \{0, 1\} \simeq 1 + 1$$

1. either exhibit $x : X$ such that $p(x) = 0$ (a root of p)
2. or else determine that p has no root.

For which infinite types X can this be done?

- **Computation.** Exhaustive search problem.
Can we exhaustively search an infinite type mechanically?
- **Topology.** Compactness problem.
Can we exhaustively search an infinite type continuously?
- **Logic.** Choice problem.
Can we prove infinite instances of choice constructively?

Our type theory

Martin-Löf Type Theory (**MLTT**)

$0, 1, \mathbb{N}, +, \times, \Sigma, \Pi, W, \text{Id}, \mathcal{U}$

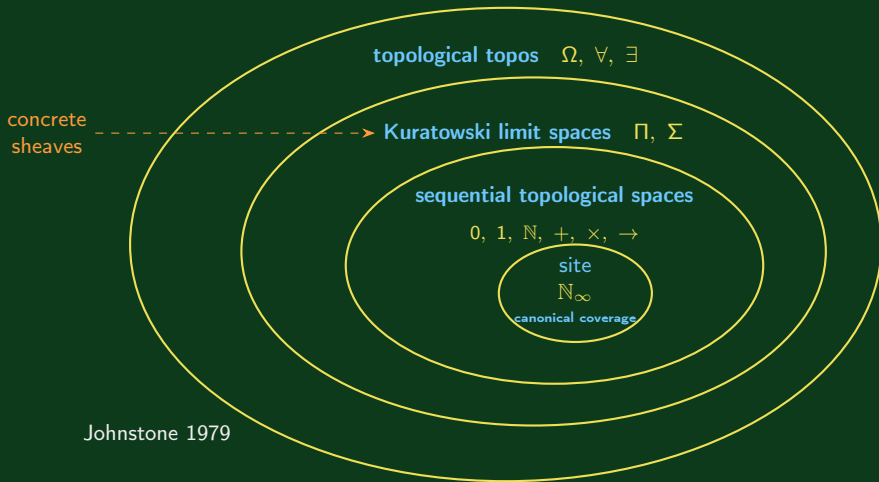
For some results towards the end, we include inductive–recursive types.

- + univalence (so in particular we have functional and propositional extensionality)
- + set quotients ($\iff$ propositional truncations + set replacement)

Many models. We reason constructively, so our results hold in all models.

- Types are sets.
- Types are topological spaces.
- Types are homotopy types.
- Types are “sets with computational structure” (realizability).
- Types are the objects of a topos, or even an ∞ -topos.

One particular model plays a guiding role



Johnstone 1979

Examples of MLTT-definable objects of the topological topos

- $\mathbb{N}$ and $2 := 1 + 1$ get the **discrete** topology.
- $2^{\mathbb{N}} := (\mathbb{N} \rightarrow 2)$ is the **Cantor space**, and $\mathbb{N}^{\mathbb{N}} := (\mathbb{N} \rightarrow \mathbb{N})$ is the **Baire space**.
- $\mathbb{N}_{\infty} := \sum_{\alpha: \mathbb{N} \rightarrow 2} \prod_{i: \mathbb{N}} \alpha_i \geq \alpha_{i+1}$ is the one-point compactification of $\mathbb{N}$.

$$\underline{n} := 1^n 0^{\omega}, \quad \infty := 1^{\omega}, \quad (n \mapsto \underline{n}): \mathbb{N} \hookrightarrow \mathbb{N}_{\infty}.$$

- $\sum_{x: \mathbb{N}_{\infty}} ((x = \infty) \rightarrow 2)$ looks like this:

$$\begin{array}{ccccccc} 0 & 1 & 2 & 3 & 4 & \dots & \\ \bullet & \bullet & \bullet & \bullet & \bullet & \dots & \bullet \ \infty_0 \\ & & & & & & \bullet \ \infty_1 \end{array}$$

This is compact T_1 but not Hausdorff.

$$\begin{array}{ccccc} \{0, 1, \dots, \infty_0\} & \cap & \{0, 1, \dots, \infty_1\} & = & \mathbb{N} \\ \nwarrow & & \nearrow & & \uparrow \\ & \text{compact} & & & \text{not compact} \end{array}$$

Mathematical expression of the problem in our system

For any given $p : X \rightarrow 2$, either find a root of p or determine there is none.

$$\prod_{p:X \rightarrow 2} \left(\left(\sum_{x:X} p\,x = 0 \right) + \underbrace{\left(\prod_{x:X} p\,x = 1 \right)}_{\iff \neg \sum_{x:X} p\,x = 0} \right)$$

- Stronger than excluded middle.
- We are making a choice.

We ask which types X satisfy this choice principle.

Definition. We call such types **compact**.

All types are compact $\iff$ global choice holds

Global choice. We can choose a point of every non-empty type.

$$\prod_{X:\mathcal{U}} \left(\underbrace{\neg\neg X}_{X \text{ is non-empty}} \rightarrow X \right)$$

- Stronger than choice, which is consistent with univalence.
E.g. Voevodsky's model of simplicial sets.
- Contradicts univalence.
- But there are plenty of compact types in MLTT and HoTT/UF.
- The ones we are able to construct are all automatically equipped with well-orders.

Functions $p: X \rightarrow 2$

They classify **complemented** (aka detachable) subtypes of X .

$$X \simeq \left(\sum_{x:X} p\,x = 0 \right) + \left(\sum_{x:X} p\,x = 1 \right).$$

$$\begin{array}{ccc} Y & \longrightarrow & 1 \\ \text{complemented} \downarrow & & \downarrow 1 \\ X & \xrightarrow{p} & 2 \end{array}$$

- **Topological topos.** They classify **clopen** subspaces.

Totally separated types

Recall definition. A type X is called **compact** if

$$\prod_{p:X \rightarrow 2} \left(\left(\sum_{x:X} p\,x = 0 \right) + \left(\prod_{x:X} p\,x = 1 \right) \right).$$

This definition is **not good** unless there are plenty of maps $X \rightarrow 2$.

(Consider $X := \mathbb{R}$ as bad example.)

Definition. A type X is called **totally separated** if

(Boolean Leibniz principle)

$$\prod_{x,y:X} \left(\prod_{p:X \rightarrow 2} p\,x = p\,y \right) \rightarrow x = y.$$

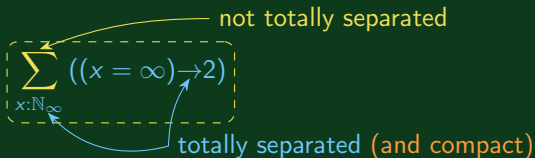
In the topological topos. The clopens separate the points.

(Topological notion with the same name.)

Some facts

1. Totally separated types are **sets** (elements can be equal in at most one way.)
2. They form an **exponential ideal**, and more generally are closed under Π , and are closed under $+$, $\times$, retracts, and include $0, 1, 2, \mathbb{N}, \mathbb{N}_\infty$ and all **discrete types** (those with **decidable equality**).
3. They are **not** closed under Σ in general.

Example. In the topological topos, the following type is **not** totally separated.



In type theory, from the assumption it is totally separated we conclude $\neg\neg$ WLPO.

Some facts continued

4. Define the **simple types** to be the smallest collection of types including $0, 1, \mathbb{N}$ and closed under $\times, +, \rightarrow$.

The **simple types** are all totally separated by (2) above.

5. **In the topological topos**, a **closed subtype** of a **simple type** is compact in the above type-theoretic sense iff it is compact in the topological sense. LMCS'2008

In this case the inclusion is a section, and hence the subtype is itself totally separated.

6. Every type X has a **totally separated reflection**, given by the image of the evaluation map

$$\begin{aligned} X &\rightarrow ((X \rightarrow 2) \rightarrow 2) \\ x &\mapsto \lambda p. p x, \end{aligned}$$

which is compact iff X is.

Counter-example to compactness

- The set $\mathbb{N}$ of natural numbers “fails” to be compact.
- The compactness of $\mathbb{N}$ amounts to Bishop’s LPO. (Limited Principle of Omniscience)

More precisely, the compactness of $\mathbb{N}$ is independent of MLTT.

- False in realizability models (finding roots is not computable)
in topological models (finding roots is not continuous)
- True in the model of classical sets (by the axiom of excluded middle)

Probably the simplest infinite example

$$\mathbb{N}_\infty := \sum_{\alpha:2^\mathbb{N}} \prod_{i:\mathbb{N}} \alpha_i \geq \alpha_{i+1}$$

This is the type of (non-strictly) decreasing binary sequences.

$$\begin{aligned}\underline{n} &:= 1^n 0^\omega \\ \infty &:= 1^\omega\end{aligned}$$

Theorem (JSL'2013)

The type $\mathbb{N}_\infty$ is **compact** in our sense.

We have an embedding

$$\begin{aligned}\mathbb{N} &\hookrightarrow \mathbb{N}_\infty \\ n &\mapsto \underline{n}\end{aligned}$$

This was originally proved in a weaker system, namely Gödel's T.

Proof sketch

with the difficult part omitted

- Given $p: \mathbb{N}_\infty \rightarrow 2$ (not assumed to be continuous), define a binary sequence x by

$$x_n = \min(p\,0, p\,1, \dots, p\,n).$$

Formula for the infimum of the roots.

- This is clearly decreasing, so $x: \mathbb{N}_\infty$.
- Now we check whether $p\,x = 0$ or $p\,x = 1$:
 - If $p\,x = 0$, then we've found a root (in fact the smallest one).
 - If $p\,x = 1$, then $p\,y = 1$ for all $y: \mathbb{N}_\infty$, and so there is no root.
(In this case $x = \infty$.)
(This is easy classically, but less so constructively.)
In the pub $\mathbb{N}_\infty$ there is a person $x: \mathbb{N}_\infty$ such that if x drinks, then everybody drinks.

Some consequences

(decision procedures)

1. For every $p: \mathbb{N}_\infty \rightarrow 2$ either $\prod_{n:\mathbb{N}} p\ n = 1$ or $\neg \prod_{n:\mathbb{N}} p\ n = 1$. (JSL '2013)

Quantification over $\mathbb{N}$ rather than $\mathbb{N}_\infty$! An instance of **WLPO** that just holds.

Bishop's principle of Weak Limited Omniscience is

$$\prod_{p:\mathbb{N} \rightarrow 2} \left(\left(\prod_{n:\mathbb{N}} p\ n = 1 \right) + \neg \left(\prod_{n:\mathbb{N}} p\ n = 1 \right) \right),$$

which is independent of MLTT.

2. (a) For any given $f: \mathbb{N}_\infty \rightarrow \mathbb{N}$, we can decide whether it is **not** continuous. (MSCS '2015)
(b) There is some **discontinuous** $f: \mathbb{N}_\infty \rightarrow \mathbb{N}$ iff **WLPO** holds.

Some applications of the compactness of $\mathbb{N}_\infty$

1. Dag Normann & William Tait.

Springer '2017

On the Computability of the Fan Functional

They use the system $\mathcal{T}$ compactness of $\mathbb{N}_\infty$ to fill a gap in an unpublished but widely circulated 1958 manuscript by Tait.

2. Cécilia Pradic & Chad E. Brown.

arXiv '2019

Cantor–Bernstein implies excluded middle

(Also implemented in Rocq.)

Compact sets in our type theory

1. 0 , 1 and $\mathbb{N}_\infty$ are compact.
2. If X and Y are compact then so are $X + Y$.
3. If X and Y are compact then so is $X \times Y$. (Baby Tychonoff.)
4. If X is a compact type and A is a family of compact types indexed by X , then its disjoint union $\sum_{x:X} A_x$ is a compact type.
5. If furthermore
 - (a) the type X has at most one element (it is a proposition in the sense of HoTT/UF), and
 - (b) we have a function that picks an element of A_x for any given $x : X$,then the cartesian product $\prod_{x:X} A_x$ is compact. (Micro-Tychonoff.)

Can we prove full Tychonoff?

No.

- E.g. the compactness of the Cantor type $2^{\mathbb{N}}$ is independent.

1. **True** in the topological topos

(it is the Cantor space)

2. **False** in Hyland's effective topos

(Kleene tree to blame)

(realizability topos over Kleene's K_1)

3. **True** in the Kleene–Vesley topos

(realizability over Kleene's K_2)

Perhaps amazingly, the toposes (1) and (3) have the **same** simple types.

More precisely, the full subcategories on the objects that arise as the interpretation of the simple types are equivalent.

Building more compact types

- The compact types we have constructed so far are all **well-ordered**:
 1. $0, 1, \mathbb{N}_\infty$
 2. $X + Y$ everything in X smaller than everything in Y
 3. $X \times Y$ lexicographic order
 4. $\sum_{x:X} A_x$ lexicographic order
- But we can't get very high ordinals with just the above.
- This is what we address next.

Well-ordered type

A type X equipped with a proposition-valued relation $<$ such that

1. The relation $<$ is transitive. (transitivity)
2. If two points have the same predecessors, then they are equal. (extensionality)
3. The order $<$ satisfies transfinite induction: (well-foundedness)

$$\left(\prod_{x:X} \left(\prod_{y:X} y < x \rightarrow P y \right) \rightarrow P x \right) \longrightarrow \prod_{x:X} P x.$$

Some consequences:

- X is automatically a set by (2) (equality types have at most one element).
- Trichotomy $x < y \vee x = y \vee x > y$ for all ordinals is equivalent to LEM.
- But there are plenty of trichotomous ordinals without assuming LEM.

The large type of all small ordinals

We also refer to well-ordered types as **ordinals**, and collect them in a (large) type **Ord**.

Univalence implies that the type **Ord** of **small** ordinals

1. is itself a **large** ordinal, and
2. has suprema of **small-indexed** families.
(We'll discuss this further later.)

Ordinal sums

1. Let $\langle \alpha \rangle$ denote the underlying type of a well-ordered type $\alpha : \mathbf{Ord}$.
2. Given $\alpha : \mathbf{Ord}$ and $\beta : \langle \alpha \rangle \rightarrow \mathbf{Ord}$, we order the type

$$\sum_{x : \langle \alpha \rangle} \langle \beta x \rangle$$

lexigraphically:

$$(x, y) < (x', y') \quad := \quad (x < x') + \underbrace{\sum_{p : x = x'} \text{transport } p \, y < y'}_{\text{naively, } (x = x') \times (y < y')}.$$

Problem with lexicographic sums and possible solutions

- The lexicographic order is **not extensional in general**. (Can derive excluded middle.)
- It **is** extensional if
 1. The given orders have **top elements**. (Then so does the sum.)
This is very lucky because our compact ordinals do have top.
 - or 2. The given orders are **trichotomous**. (Then so is the sum.)

Suprema of families of ordinals

- Does every family $\alpha: I \rightarrow \text{Ord}$ have a supremum **constructively**?
- As far as I know, this hadn't been answered before.
Left open by Forsberg, Kraus & Xu (MFCS '21).
- Two answers, by Tom de Jong and myself independently.
Both constructions and proofs implemented in Agda by Tom.
- Mine is as follows.

The **supremum** is just the **image** of the function

$$\sum_{i:I} \langle \alpha_i \rangle \longrightarrow \text{Ord}$$
$$(i, x) \longmapsto \alpha_i \downarrow x.$$

Although **Ord** is large, the image is small (**assuming quotients**).

Suprema of families of ordinals continued

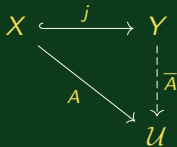
Corollary If I is compact and each α_i is compact, then so is $\sup_{i \in I} \alpha_i$.

Because $\sum_i \alpha_i$ is compact and images of compact types are compact.

Extending families of types

We need a further ingredient.

General situation:

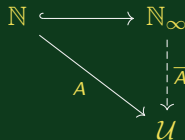


A : type family.

$\overline{A}$: extension of A along an embedding j .

$\mathcal{U}$: type universe.

Interested in:



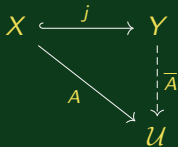
canonical embedding.

Want: If Ax is compact for every $x : X$, then $\overline{A}y$ is compact for every $y : Y$.

Because then: If Y is also compact, then $\sum_{y:Y} \overline{A}y$ is compact too.

Family extension problem

MSCS '2021, "Injective types in univalent mathematics"



Consider

$$j^{-1}(y) := \sum_{x:X} j\,x = y.$$

This type has at most one element
(because j is an embedding).

Smallest solution (left Kan extension):

$$\underline{A}\,y := \sum_{(x,-):j^{-1}(y)} A\,x$$

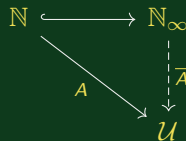
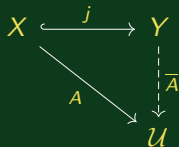
Largest solution (right Kan extension):

$$\overline{A}\,y := \prod_{(x,-):j^{-1}(y)} A\,x$$

It is this that works for the wish of the previous slide, by Micro-Tychonoff.

Extending families of compact types to compact index sets

Special case of interest:



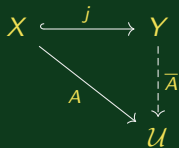
$$\bar{A}_y := \prod_{(x,-): j^{-1}(y)} A_x \quad (\text{product of a family with at most one element}).$$

Theorem If the type A_x is compact for every $x : X$,
then the type $\bar{A}_y$ is compact for every $y : Y$.

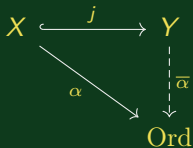
Corollary If additionally Y is compact, then so is $\sum_{y:Y} \bar{A}_y$.

But this is not enough

The above says that type universes $\mathcal{U}$ are injective.



$$\bar{A}(y) = \prod_{(x, -): j^{-1}(y)} A x$$



Theorem. The type of (topped) ordinals also is injective.

Proof. That of the injectivity of a universe + additional construction of the order + checking it works.

We need to order $\bar{\alpha}(y)$.

We define, for $\mu, \nu : \langle \bar{\alpha} y \rangle$,

$$\mu < \nu := \sum_{\sigma: j^{-1}(y)} \mu \sigma < \nu \sigma.$$

Next

Using the previous machinery.

1. Compact ordinals induced by an **inductive** type of Brouwer ordinal expressions.
2. A generalization to an **inductive-recursive** universe à la Tarski of compact ordinals.

Brouwer ordinal codes

A type B inductively defined by **constructors**

$Z : B$	“zero”
$S : B \rightarrow B$	“successor”
$L : (\mathbb{N} \rightarrow B) \rightarrow B$	“limit”

A W type of countably branching trees.

Four interpretations of Brouwer codes as ordinals

0) standard interpretation

$$\begin{aligned}\llbracket Z \rrbracket_{\text{sup}} &= 0 \\ \llbracket S b \rrbracket_{\text{sup}} &= \llbracket b \rrbracket_{\text{sup}} + 1 \\ \llbracket L b \rrbracket_{\text{sup}} &= \sup_{i:\mathbb{N}} \llbracket b_i \rrbracket_{\text{sup}}\end{aligned}$$

1) trichotomous interpretation

$$\begin{aligned}\llbracket Z \rrbracket_{\Sigma} &= 0 \\ \llbracket S b \rrbracket_{\Sigma} &= \llbracket b \rrbracket_{\Sigma} + 1 \\ \llbracket L b \rrbracket_{\Sigma} &= \sum_{i:\mathbb{N}} \llbracket b_i \rrbracket_{\Sigma}\end{aligned}$$

2) compact interpretation

topped

$$\begin{aligned}\llbracket Z \rrbracket_{\overline{\text{sup}}} &= 1 \\ \llbracket S b \rrbracket_{\overline{\text{sup}}} &= \llbracket b \rrbracket_{\overline{\text{sup}}} + 1 \\ \llbracket L b \rrbracket_{\overline{\text{sup}}} &= \sup_{i:\mathbb{N}_{\infty}} \overline{\llbracket b_i \rrbracket_{\overline{\text{sup}}}}\end{aligned}$$

3) compact totally separated

topped

$$\begin{aligned}\llbracket Z \rrbracket_{\overline{\Sigma}} &= 1 \\ \llbracket S b \rrbracket_{\overline{\Sigma}} &= \llbracket b \rrbracket_{\overline{\Sigma}} + 1 \\ \llbracket L b \rrbracket_{\overline{\Sigma}} &= \sum_{i:\mathbb{N}_{\infty}} \overline{\llbracket b_i \rrbracket_{\overline{\Sigma}}}\end{aligned}$$

Overline means extension to $\mathbb{N}_{\infty}$ by injectivity.

Assuming excluded middle

standard

$\llbracket b \rrbracket_{\text{sup}}$

$\leq$

$\llbracket b \rrbracket_{\Sigma}$

trichotomous

$\mid \wedge$

$\mid \wedge$

compact

$\llbracket b \rrbracket_{\overline{\text{sup}}}$

$\leq$

$\llbracket b \rrbracket_{\overline{\Sigma}}$

compact totally separated

In the next slide we see what happens constructively here.

Why do we need excluded middle?

Because $(-)+1$ is monotone $\iff$ excluded middle holds.

Theorems

Even better: every complemented subset is either empty or has a least element.

The ordinal $\llbracket b \rrbracket_\Sigma$

- is discrete, and moreover trichotomous
- is a retract of $\mathbb{N}$
- so countable
- not compact unless LPO holds

The ordinal $\llbracket b \rrbracket_{\overline{\Sigma}}$

- is compact
- is a retract of $\mathbb{N} \rightarrow 2$
- so totally separated
- not countable unless LPO holds
- not discrete unless LPO holds

-
- There is an order-preserving-reflecting embedding $\llbracket b \rrbracket_\Sigma \hookrightarrow \llbracket b \rrbracket_{\overline{\Sigma}}$ whose image has empty complement.
 - LPO $\Rightarrow$ this embedding is a bijection $\Rightarrow$ WLPO.
 - In models: the embedding doesn't have a computable/continuous inverse.

Illustration

Two manifestations of the ordinal $\omega + 1$.

$$\mathbb{N} + 1 \hookrightarrow \mathbb{N}_\infty$$

$$n \mapsto 1^n 0^\omega$$

$$* \mapsto 1^\omega$$

$\mathbb{N} + 1$ is

- discrete
- compact iff LPO
- countable

The map $\mathbb{N} + 1 \hookrightarrow \mathbb{N}_\infty$

- is a bijection iff LPO,
- but its image always has empty complement.

$\mathbb{N}_\infty$ is

- compact, totally separated
- discrete iff WLPO
- countable iff LPO

Every decreasing sequence is of one of the forms $1^n 0^\omega$ and 1^ω . Constructive taboo.

There is no decreasing sequence other than $1^n 0^\omega$ and 1^ω . Just true.

Generalizing and extending the above

In order to successively construct more and more compact ordinals in stages,

1. in the **above**, we take suprema/sums of $\mathbb{N}_\infty$ -indexed families of ordinals, and
2. **next** we allow α -indexed families, where the compact ordinal α is constructed in previous stages.

That is, we move from ω -branching trees to ordinal-branching trees.

Universes à la Tarski of compact ordinals

We define

$E: \mathcal{U}_0,$ (infinitary) ordinal codes or expressions

$\Delta: E \rightarrow \text{Ord}^\top$ discrete and trichotomous interpretation

by induction–recursion.

After that, we define, by recursion,

$K: E \rightarrow \text{Ord}^\top$ compact (totally separated?) interpretation

We define E , using Δ , inductively by the following constructors:

$$\begin{array}{l} \ulcorner 1 \urcorner: E \quad \left| \quad _ \ulcorner + \urcorner _: E \rightarrow E \rightarrow E \quad \left| \quad \ulcorner \Sigma \urcorner: (e: E) \rightarrow (\langle \Delta e \rangle \rightarrow E) \rightarrow E \right. \\ \ulcorner \omega + 1 \urcorner: E \quad \left| \quad _ \ulcorner \times \urcorner _: E \rightarrow E \rightarrow E \right. \end{array}$$

Tarski universes continued

We simultaneously define

$$E: \mathcal{U}_0, \quad \Delta: E \rightarrow \text{Ord}^\top,$$

and then we simultaneously define

$$K: E \rightarrow \text{Ord}^\top, \quad \iota: (e: E) \rightarrow \langle \Delta e \rangle \rightarrow \langle K e \rangle,$$

$$\iota\text{-emb}: (e: E) \rightarrow \text{is-embedding}(\iota e).$$

Recall the E constructors:

$$\ulcorner 1 \urcorner, \ulcorner \omega + 1 \urcorner, \ulcorner + \urcorner, \ulcorner \times \urcorner, \quad \ulcorner \Sigma \urcorner: (e: E) \rightarrow (\langle \Delta e \rangle \rightarrow E) \rightarrow E.$$

We define $\Delta \ulcorner \text{brocoli} \urcorner = \text{brocoli}$ recursively.

K is defined as Δ except that rather than

$$\Delta(\ulcorner \Sigma \urcorner e A) = \sum_{x: \langle \Delta e \rangle} \Delta(A x) \quad \text{we define} \quad K(\ulcorner \Sigma \urcorner e A) = \sum_{x: \langle K e \rangle} K(\overline{A} x)$$

where $\overline{A}$ extends A along ι using the injectivity of $\text{Ord}^\top$ as in the next page.

Tarski universes concluded

$$\begin{array}{ccc} \langle \Delta e \rangle & \xleftarrow{\iota e} & \langle K e \rangle \\ \downarrow A & \searrow & \downarrow \overline{A} \\ E & \xrightarrow{K} & \text{Ord}^\top \end{array}$$

Theorem All compact well-ordered types we constructed above have the stronger property that every complemented subset is either empty or has a least element.

This is a property that well-ordered types don't have in general in a constructive setting.

The End